

DESAFIO



Exfiltração

100

Um grupo de Black Hats executou um APT e exfiltrou arquivos de uma empresa, porém não estão conseguindo decifrar tais arquivos.

Diante do exposto, descubra quais arquivos foram exfiltrados e tente decifrá-los obtendo a flag para pontuar.



PONTUAÇÃO BÔNUS PARA OS PRIMEIROS A RESOLVEREM



+70 pts



+50 pts



+30 pts



ctf.pcap

Bandeira

Enviar

Filtrando por http.request.uri

Conseguimos localizar um arquivo chamado ctf.priv

Se tratando de uma chave privada.

-----BEGIN PGP PRIVATE KEY BLOCK-----

```
lQVXBGJ6zcIBDACfpQezNkw0b4gvDVDz4MlFM1j3WAgQMouCMMgdQWfMhoVrf/5c
M8gtWT8m0z1GHsGPzLLiPRI71si/cJPM56St8BHiZrSabkuDfvTFF9ZtZdXeZnvL
Vy57b0lNe8ux1joSGak/nIN2ugLdVq9RSOTGhIBjp9mI8bo3qk92iqQYc2ogT1ML
abT/Mzp7Sdi6A1sdVsCb0KNL+fp+wcMZKmQ3sT0eUbbjhb7hSXQjxLydWxA8ANza
```

dy0oEPCy6T3V0fsitfxA//nNEpfKya92KoTSiGZ9obWh4VRqZ3t8pFmHnRwNGaru
jL/7ju0FupeNYDJC04hJbjDtomfc3qwCErVhjodiFTzYE0d3WngEtNmp7TaJe1ph
3wqVPJ2VjLVMBgq7gIoGc5/r3goxAh7Po1pgg5b60Tn+KMLeJErIOJrtlhe/Gsw1
+pWdjA3FWVCy00jym5fT2v5H7CBmZluJtj3ok8HY1g70R2MByABMkRSBTs+g24xM
qp9iuYiSeGpZ1/cAEQEAAQAL93DfPZJ3zNDGCoM55wKG68cBhuVaWfkbY4eo9d05
aqowciSPWkfWVLG7Z+sLb0ckd7Y0BEXw5/Nm6IhyEA3PuZ9BHWP10bv2P225VVYE
hJbUaWHVE7Gw8ixR1KHZ5RDXZjd17UQMZOaMnXng0B0odhf1luDey+8G+aJhx80I
OP0QnPg7HXv0QQQuNCr3C0R5GvKjY4pbgRk2pOT6GfaxAipeHyDQpX44Eb03MeYY
M0MOH2xNL9lGaabqU0QrpIyI60jCjKPO5mzBFwuNmuXmsh5wOpPts3BK4jXdcy1s
CTI6tit66GKItgkea7mifYkPGZ9b0165h1ArDaIkYzH6M+QbY+a1k7bK0e3ECIZZ
EfJPiHZH5Y892yH7TKbIwAFVovVRArQR8B9ZowKD14HfE+seotTga4LLgOBtRH8a
k61y/6o5mifNaNS82AvN2cOP9KniY+1h3KXXnxyJaV10xMdRsQRIZj9eo5H1D/LI
UwvLuA9uYheihSJ/1qx28ywJBgDJIC1u7lZKUc38nAp1nJ2NnsFpMg4ElhSdFfe3
9fhZaUvBkvaNQhXJKBPGwId8E9v1nE64dnx3QKEEu3ieM55UozlrxDX2zI3bfbxJ
RUgvxKhNvszX4KKjEfkyLIL3pbGzHwIJyHAjD6yk8CXY4qtPGdWfa2muo0m3tTb1
HHj4KBABRVf3A0xtE9wgVI59l6ugA040AqWYyffo+jemx6mYT8mH89UIzv7sBNB+
2rnfq5BUmsiOrUD5i+XepTPDeCUGAMszkw/41K8WWGjgbk2RCGihvhIbG83iHnHBT
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EdNEv4Ve6IXldKc17Xc4hTiaeDi2SDBbPecou0n9bYgeJ2H4mk6S6c0Xi5X9C/iE

yiVsLMHoYGu+CkWuy+6/U/jHtQ6l1DF+7uGzYr0VH4s2IXp0A3UdhWm/pDtUG2P
mq85wLf/XmCKg6G0CKtoCMR7ncalJvaA5J00EmNmdGZpYXAgPGZpYXBAY3RmPokB
1AQTAQoAPhYhBDLzi4wdPgmadpKYi9Dkm/cobHYeBQJies3CAhsDBQkDwmcABQsJ
CACCBhUKCQgLAGQWAgMBAh4BAheAAAoJENDkm/cobHYeR+AL/1qm4qaf1Zfp2wJP
1JexF5FVuqbxiz6dewLqPro3FilSs5r/0s958PIwfVQY3qUquSEl/mMRCizSNYvp
6qC9DLj94icgw5Dqa0FVLD1A1KdteW4W6r/MVoGxgh2ZvfamQitMXLj7i0yVkySc
+V0X/QrQ30lcPGQnmk0FrpFYe0NbI8PRhuk2tscostoq/VzSKY3xHkJYkb4uxqWF
CrwSJk4ktqJ5ABwmYc5TMIByifr8f+hpjsCc6Qi+IsOga38G1HQr/5GxVJe3D+2f
TDcFyHcH/tRBK0cjmJaoHz8q/r2fHsWUAFBXHdIRRJkwG8Ml+HGDKiWVK0eRgP8s
dfgVG6zN7/RLLkaeKcmbPIEWzzFKVakaDSD9eFxl1eA2EupuoPXnZ3Ase+J0hNJ
VCe6hNqg8sDrimL8+5leIzUtW72qlUQU+IkZlYgY554xdvq3S/KbYrESOGTxw9xo
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0QI54wzv2YSrSgsK0sNbi1ZwqyDQJuK+5HUy/bx17uQnREvx+FudY84r90qTeYQh
6Uq1iYPAYeU9YfrpPuGLjFCVyXI8ftFW1YSntE8kbguFemZvcnpC+60jTeIIdxmu
dXv1h/iYj+f7SsBJKPPNyWQULJHc5MeNrHpattSbcrg4mgHD+u0HafgUd0gwKYx
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j09Q+9v6/Q2ICH1eLBdgevvp8yqdJtZztQCwGZSPz60StIxK2gAurkEfirBTg0wL
nVwG0//K8YQSHgd1zBpTXLQ6UXvzVm36BN8yXPDEkIX125ea+BRMEivTZypc4tw4
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wez2t+mJ0Tn06GAFES8dC4CxFNpaxIyb4KCCu1CFvWrE4T+KNDFfZdThP6ckA1DP
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4gAoHvhL90Xg0qYTVN7MiT6RYFWyEaiwC344SIFHYbhdQEPmzN++21sRDrdMLhyR
JXj4MizJ78pbnRS02glcfC988VckVqFMf0xbW0XZWnGC8/udVPsMBM0GeiKI1y7i

9tfOZD+MKSaL9E3EzkdCINm22kCMPe+d6BvJqjo10S/yGLif/aPsrc9PwZAE2Nho
jItJ16JsUdMo/u4Y6QUqqTNWND7s91aN6PWdabbEQ7ntps/vK6w+aoQWgyUoSrrj
NBshy1pNmMy+Yw+Ses+JSS3vGAYsPRMoB4wwBPZwSnuqGUdb0eKcxw0jUiRGznd1
Cz6AlWB1YU/VWzOgaC/249g0VtGVsvxPvWn8cGvL3ur5acE2Nt8svh85zPFQ0bJ7
HB/GbHCLUeiZ4M4M+0sZxDtYPYthxuQrQfQbPFsNfdSHvR98rHZCIxc4Z3hjoy20
Mb6+n9LtLVUGANIyU0V159NE103NKqCRuLWWJJqKqNP17cR9JTG2KvuXE7t+WArr
NJpkKPNUeIoaR4ibSF/oIBYVAFcOTyyCWqbnUxFVQRsrjbEaobYraYQ+WnLuc6Rw
nVTtMHnwsRI/VFL8N+0tQ0Mro/vIUtr8vGVXRPT6Un9MUSgryZHRiOVfsDKUA0Je
76n1KbmrDNkYZ/vIturz/HYTKFdDk3VI5Y06vJFKsoChlJDbuVVDjnwjHwqjkeQV
zbmUsqBfhf0cPQYA/o2ipiYzuo40lkC0u8zsuA9hPuNj21Ja846UZFT2wiqwXVnD
MG5xfqBDdSXlQz091TjtYtgmUARFzGrX5exG2HzHBHa9uE9d+C7ondwNZrR7IIVN
iM8VRpYhvtY1MORRIL3sTLFw4W8SDPJb33qT79SFx5GibnzeWoNs4j6QzGcBi+Er
pkFEzucMdafIMiUFSOfu3eEBPxlnZsI9ERV7tpHskU0KjQqpxIw+AekdaX7lzlFA
QiaWywzbeJda6zX7BgDdwTiDw7oeD0SGeAFqTUBnaS3egRFh72DBqbXH12Iydc1U
L4tRzJmxLi+GU5lTGsRQ6KLphpPTGlorekxKJDAjBvqLlqY0zmQiFhgjVPWNKiQI
4rSMN1vnc19S0rVAWATHqX2/rk0TXXI9cZZl3yiU78vAB5h6WzLyvBC8xXi03z7I
srZa/VhX0i52RSP+GmDTNX9nAyr1FsZaPa6Ng9ALJrbF+z3TT22cTEjUXt8s6IL9
4WRQ6tGPQok9/48bfIbacokBvAQYAQoAJhYhBDLzi4wdPgmadpKYi9Dkm/cobHYe
BQJies3CAhsMBQkDwmcAAAOJENDkm/cobHYe6LsL/2DLnp/jiVp4GCLeZ8eYilNI
8M4yuLQdXk8PUCp3hayU3ezMhzVgDJPMZpze88pQdJRcdleQEzr2UJZQWIyoaXCA
ZIUgoOPGh/HYPmlgRqXHsvLsvf3EJj1bpG7yC5kbZ0pm2Ax1STE9youiE/6UJ2R8
QnydAenroEWPAf00A1sU4Eb81GkK9y12pxZsi0m/Pss+FZV+rFVxndGvQIEs4b1F
/JLHLPACpLtBWeSczFLfLV1LOSsHJwcG/Bt+gq6qrhfyEgRPQmTJP7fchSBm2HwM

```
Ik1JzE1wkLLiJif57Zxeryiew82IvFvn/A1Q3+cPxafn5no0TpQevUZy4zF0wxM2  
b7M0hCh56b34pk35hfENoIeJhEON7jX8Ii5CV/mTgb3DDyTiWdHwrK6ZJevAtLj1  
KmW7BGe0b4sw8yFUodnhme1A2N8cunVy2CPHS5Lja1w/NBQldEwHOy50b/zilKAU  
vBgEX8g+Wp04/n66WIVRNz7J8MDYo6h4Cqiz0qUBhQ==  
  
=qhaC  
  
-----END PGP PRIVATE KEY BLOCK-----
```

Agora o arquivo secreto.asc:

```
-----BEGIN PGP MESSAGE-----  
  
hQGMAwyfAso8RYdJAv6A9zw0odAbLOI8wnApXPzpi+blzi+EEICH0gZgYIPibfA  
NyTZoiobpG7e0KuvpJ9K9qTt/3CjVX3Jjg+yi8sNMfZ/tjNNpEJOhqGhtsezP0v  
NRk1tMYzjtYlJbv7iozIQdRGGiY92Q4U06vaEDGQWZuVN05sPsGZ/juwFzG2ULP8  
4Mc/oGGVjrqrE5NDDErsGzS9zXPLUEhlihqb79+23E8UpY195EVkyX2tex9R54s  
Aywhoj9ZPt2IIInHqGtVy/QNUcF9EWIzWpN1BoGjz5ShDj8kNctyEUxLbmtbcGvDH  
t+xaXLdphIP9WxyYDR3npYWgGLp1FSVuHvnibw8D+PTRSOu6rdTWCmsaJ3hL408f  
dLsG+zMWhEvQFIeq7fAugOZv1mgZBIMob5+0DHYW4fVyjdtlQ+2pq3BYmPf08MI6  
D+PsjRVEOWTgmW4apKuirCicKe0e006maxk7p8BzJSEZuRgpbhZGCM8W4+0kP0qG  
xKDCJM8rPIsxZuPRNCwd0kwBVnFaf0S1Pirno1p3bRjw1/6ZaE84ckEZkqT3o1Bg  
kbrJRkDl0jIJn+Ho6ATe9HWA0oE+t/Ef38wBiMjoFUV+GEGVI4hhIoxA36vK  
  
=UqCf  
  
-----END PGP MESSAGE-----
```

Colocado os arquivos no linux e feito a decodificação:

```
(root@DESKTOP-EM0E755)-[/home/cereal]
# gpg --list-secret-keys
gpg: /root/.gnupg/trustdb.gpg: trustdb created

(root@DESKTOP-EM0E755)-[/home/cereal]
# gpg --import ctf.priv
gpg: key D0E49BF7286C761E: public key "cftfiap <fiap@ctf>" imported
gpg: key D0E49BF7286C761E: secret key imported
gpg: Total number processed: 1
gpg:         imported: 1
gpg:         secret keys read: 1
gpg:         secret keys imported: 1

(root@DESKTOP-EM0E755)-[/home/cereal]
# gpg --list-secret-keys
/root/.gnupg/pubring.kbx
-----
sec   rsa3072 2022-05-10 [SC] [expired: 2024-05-09]
      32F38B8C1D3E099A7692988BD0E49BF7286C761E
uid   [ expired] cftfiap <fiap@ctf>

(root@DESKTOP-EM0E755)-[/home/cereal]
# gpg --decrypt secreto.asc
gpg: encrypted with rsa3072 key, ID 0C9F02CA3C458749, created 2022-05-10
      "cftfiap <fiap@ctf>"
gpg: Note: secret key 0C9F02CA3C458749 expired at Thu 09 May 2024 05:40:34 PM -03
FIAP{gpG}

(root@DESKTOP-EM0E755)-[/home/cereal]
# _
```

R: FIAP{gpG}